

Can Blue-Chip Talent Carry the Boat?

A College Football Recruiting vs. Wins Predictive Modeling Project

Background and Motivation

One thing I learned working with Notre Dame's Football Recruiting Department is how much recruiting contributes to a program. From landing a 5-star athlete in the 2026 class to picking up a breakout transfer from Ohio State, the impact recruiting has on what happens on the field is real and significant.

I built this project to answer a baseline question: how much does recruiting actually contribute to on-field success? Coaching, player development, and scheme all matter too, but I wanted to put a number specifically on recruiting. What is it actually worth?

The Data

I pulled 14 years of recruiting class rankings and win/loss records from the College Football Data API, covering 255 FBS programs from 2010 to 2023. The final merged dataset contained 2,765 team-season rows with each program national recruiting rank, composite recruiting points, wins, and losses for that year.

Milestone 1 — Does Recruiting Predict Wins at the Season Level?

The first question I wanted to answer was whether a team recruiting class rank predicts how many games they win that season. From the results, not really. But I needed to understand from what point of view this was.

Within college football there are various conferences with teams assigned to each one, excluding Notre Dame as an independent. Power 4 conferences consist of teams that are well funded, meaning they have the facilities, equipment, staff, nutrition, and talent acquisition to function as a powerhouse program. The problem with how I initially collected my data was that FBS teams were also included. While FBS programs are still considered Division 1 and sometimes compete head to head with Power 4 teams, the funding and playing level they are at differs drastically.

Take Arkansas State and the University of Arkansas as an example. Arkansas State is a Sun Belt team competing with other programs that get roughly identical funding and have similar facilities. The University of Arkansas is an SEC program with newly upgraded facilities and a large budget.

On top of that, being in the SEC means competing with teams that are either in the same boat or even better. A comparison between the two is not fair or reasonable. The funding is off, the player development differs drastically, and any head to head matchup would be heavily one-sided.

The takeaway is that with only a 3% win variance ($R^2 = 0.03$), it makes sense that the relationship between recruiting rank and on-field success is weak across all of FBS. Once the dataset was filtered to Power 4 programs only, the win variance jumped to 18% ($R^2 = 0.18$). Recruiting now explains roughly 1 in 5 wins. That is a real and meaningful signal.

Milestone 2 — A Cleaner Question: Game-Level Head to Head

Along with filtering out FBS schools, I also had to consider that schedules differ across Power 4 teams, which adds noise to any season-level comparison. So instead I asked a more precise question: which team would have the edge in a specific matchup based on recruiting history?

What I did was calculate each team 4-year rolling average recruiting rank to approximate current roster talent built from recent classes, then compare that differential head to head to predict a winner. This is a better approach because rather than predicting outcomes across an entire schedule, you are predicting the winner of a specific matchup based on recruiting talent.

The recruiting-only model achieved 62.7% accuracy with an AUC-ROC of 0.69. AUC-ROC measures how well a model ranks outcomes. In terms of football matchups, it tells you how often the model correctly assigns a higher probability to the actual winner. A coin flip sits at 0.50. Vegas sportsbooks operate around 0.70 to 0.72. Getting to 0.69 with recruiting rank as the only signal is a significant result.

It is important not to confuse AUC with R^2 . R^2 measures explanatory power, how much of the outcome a variable explains. AUC measures predictive accuracy, how good the model is at picking the right team. Both matter but they answer different questions.

Milestone 3 — Adding Returning Production

While recruiting is a good measure of a football program potential, one thing I had to keep in mind was how a player develops from their redshirt year to their junior year. That is where returning production comes in.

Returning production measures how much of a player previous season statistical contribution will be returning for the upcoming season. It captures experience, development, and continuity. The 5-star redshirt who only got 6 in-game reps will probably not have the advantage over a 2-star player entering their third year who started all 12 games the previous season.

When returning production was added to the model, it was the clear winner over recruiting rank. The feature coefficient was 1.23 compared to nearly zero for recruiting rank, which proves that a developed player will outperform a first-time starter in terms of predictive value. AUC improved to 0.70 with an accuracy of 64.2%, meaning 12 additional games out of 877 were correctly identified.

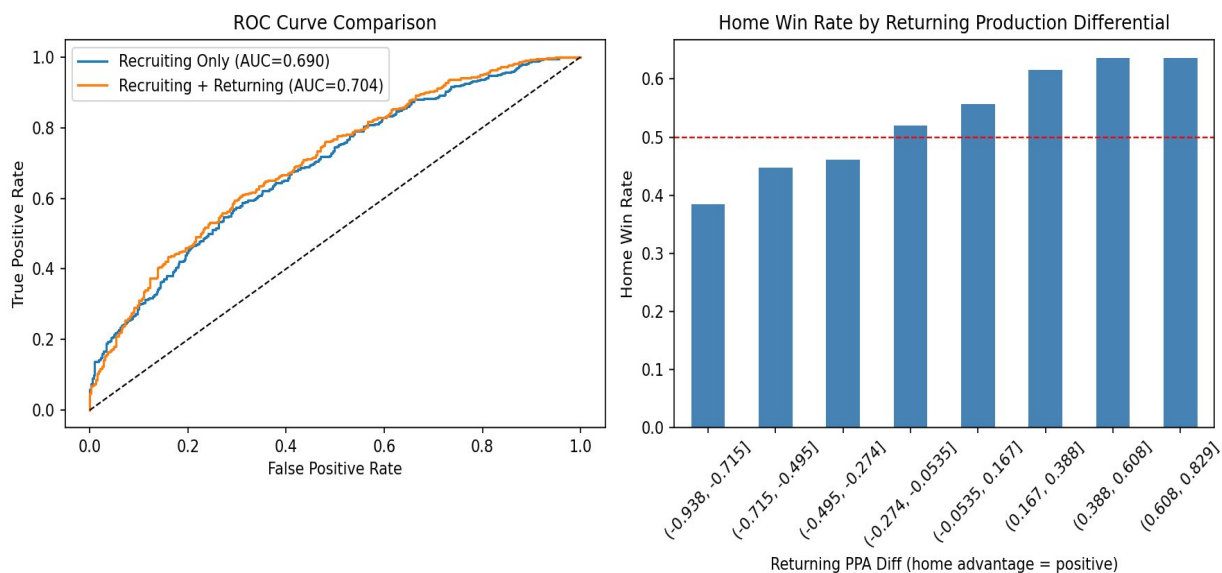


Figure 1: ROC Curve comparison between recruiting-only and recruiting + returning production models (left). Home win rate by returning production differential (right).

Summary of Results

Model	Features	Metric	Score
Season Win % — All FBS	Recruiting rank + lags	R ²	0.03
Season Win % — Power 4	Recruiting rank + lags	R ²	0.18
Game Outcome — Recruiting only	4yr avg rank differential	AUC-ROC	0.69
Game Outcome — Full model	Recruiting + returning production	AUC-ROC	0.70
Vegas Benchmark — Model	2021-2023 test set	Accuracy	64.1%
Vegas Benchmark — Vegas	2021-2023 test set	Accuracy	71.0%
Model on disagreements	Games where model $\neq$ Vegas	Accuracy	37.5%

Milestone 4 — The Vegas Benchmark

The final comparison was against Vegas point spreads. Vegas lines are constructed through a market process where various bettors, analysts, and insiders place money on games, which means that whenever a player is injured the bets move within minutes.

Within the 2021-2023 test set, Vegas predicted the winner 71% of the time while the model hit 64%. That gap makes sense because Vegas considers injuries, coaching patterns, past game performances, weather, home field advantage, and years of market refinement. On top of that, there were 243 games where the model and Vegas disagreed, with the model being right on only 37.5% of them.

Recruiting and returning production are not the complete predictors of college football outcomes. Vegas already prices in the smaller factors that contribute to those outcomes. The only way to build a model that approaches Vegas accuracy would be to incorporate efficiency ratings, recent performance trends, injury data, and live updating as new information comes in before kickoff.

What I Learned

The biggest thing I learned was that experience beats potential in the short term. Recruiting rank tells us what a program wants their roster to look like in the next 2 to 3 years. Returning production is the construction of the roster the recruiting department envisioned.

I also learned a lot about how to filter data properly. The comparison between FBS schools and Power 4 schools shows clearly in the win variance of 0.03: weak and not relevant. The right schools need to be compared to get accurate results. When the comparison is fair, the data tells a much cleaner story.

What Comes Next

This project opens up a larger question I want to explore: how much do different departments within a college football program contribute to on-field success? Coaching efficiency, strength and conditioning, NIL investment, portal strategy. There are a lot of variables left to evaluate. I am curious to see what other questions come up as I dig deeper into what actually separates good programs from elite ones.

GitHub: github.com/mcisnerosy/recruiting-class-analyzer